



Free and Forced Convection

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1. Theory

Background

Heat transfer is a fundamental aspect of thermal engineering, involving the transfer of thermal energy between physical systems. There are three primary modes of heat transfer: conduction, convection, and radiation. Convection is particularly important in industrial and engineering applications, where heat transfer between surfaces and fluids is analyzed for both free (natural) and forced convection scenarios.

In natural convection, the movement of fluid is driven by density gradients caused by temperature differences. This process occurs without external forces, relying solely on buoyancy effects. In contrast, forced convection involves external mechanisms, such as fans or pumps, to enhance fluid motion, increasing the heat transfer efficiency.

This experiment investigates the heat transfer coefficient for both free and forced convection for different geometries: a flat plate, pipe bundle, and fins.

Governing Equations

1. Heat Transfer Rate:

The heat transferred to the fluid, Q , is given by:

$$Q = \dot{m} \Delta h$$

where:

$\dot{m}$ is the mass flow rate, calculated as:

$$\dot{m} = \rho v A$$

ρ is the air density, v is the average velocity, and A is the cross-sectional area of the duct.

Δh is the enthalpy variation, which, for air, can be simplified as:

$$Q = \dot{m} c_p \Delta T$$

where c_p is the specific heat capacity at constant pressure, and ΔT is the temperature difference between the inlet and outlet.

2. Efficiency:

The system efficiency, η , accounts for energy losses and is defined as:

$$\eta = Q / P_{el}$$

where P_{el} is the electrical power supplied to the heaters.

3. Heat Transfer Coefficient:

The heat transfer coefficient, h , relates the heat flux to the temperature difference:

$$Q = h A \Delta T_{avg}$$

where ΔT_{avg} is the temperature difference evaluated as the Log Mean Temperature Difference (LMTD):

$$\Delta T_{LMTD} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$$

For small variations, this can be simplified using the arithmetic average.

4. Dimensionless Numbers:

Reynolds Number (Re), Determines the flow regime:

$$Re = (\rho v L) / \mu$$

where L is the characteristic length, and μ is the dynamic viscosity.

Nusselt Number (Nu), Relates heat transfer to the flow:

$$Nu = (h L) / k$$

where k is the thermal conductivity. For laminar flow over a flat plate:

$$Nu = 0.664 Re^{0.5} Pr^{1/3}$$

where Pr is the Prandtl number.

Brief Description of the Experiment

The experimental setup involves an air duct where heated surfaces (flat plate, pipe bundle, and fins) are tested under natural and forced convection. The apparatus includes a control and display unit, thermocouples for temperature measurement, and a fan to regulate air flow.

The procedure involves:

- Heating the test geometry using resistive heaters.
- Measuring inlet and outlet temperatures and the volumetric flow rate.
- Calculating the heat transfer rate and heat transfer coefficient.
- Comparing experimental results with theoretical predictions, particularly for the flat plate, using the Nusselt number equation.

This experiment elucidates the impact of geometry and flow regime on the heat transfer coefficient and system efficiency.

2. Results

Data Tables

The 2 tables below summarize the experimental data and recalculated parameters for each geometry under free and forced convection. It includes the measured surface temperature (T_{surf}), inlet and outlet air temperatures (T_{in} and T_{out}), airflow velocity (w), and supplied power (P_1). Calculated values include temperature difference (ΔT), heat transfer rate (Q), efficiency (η), Log Mean Temperature Difference (LMTD), and heat transfer coefficient (h).

Geometry	Area (m ²)	Convection Type	T _{surf} (°C)	T _{in} (°C)	T _{out} (°C)	Velocity (m/s)	Power (W)
Flat Plate	0.014	Free	62.00	20.90	21.80	0.11	82.10
Flat Plate	0.014	Forced	45.20	19.80	20.60	2.91	82.10
Pipe Bundle	0.098	Free	59.70	21.50	28.20	0.17	79.00
Pipe Bundle	0.098	Forced	25.00	19.80	21.20	2.27	79.30
Rectangular Fins	0.140	Free	61.10	20.90	28.20	0.17	80.00
Rectangular Fins	0.140	Forced	31.70	19.80	21.10	2.65	79.50

Table 1: Experimental Data

Geometry	Convection Type	Delta T (°C)	Q (W)	Efficiency (%)	LMTD	h (W/m².K)
Flat Plate	Free	0.90	1.67	2.04	40.65	2.94
Flat Plate	Forced	0.80	39.31	47.88	25.00	112.31
Pipe Bundle	Free	6.70	134.62	170.40	34.74	39.54
Pipe Bundle	Forced	1.40	375.60	473.65	4.46	858.68
Rectangular Fins	Free	7.30	209.53	261.91	36.43	41.08
Rectangular Fins	Forced	1.30	581.65	731.64	11.24	369.72

Table 2: Calculated Parameters

Detailed Computations

1. Temperature Difference (ΔT):

The difference between outlet and inlet air temperatures:

$$\Delta T = T_{\text{out}} - T_{\text{in}}$$

2. Heat Transfer Rate (Q):

$$Q = \rho \cdot w \cdot A \cdot c_p \cdot \Delta T$$

where:

$\rho = 1.2 \text{ kg/m}^3$ (air density),

A = duct cross-sectional area (varies by geometry),

$c_p = 1005 \text{ J/(kg} \cdot \text{K)}$ (specific heat capacity of air).

3. Efficiency (η):

$$\eta = (Q / P_1) \cdot 100$$

4. Log Mean Temperature Difference (LMTD):

$$\Delta T_{\text{LMTD}} = [(T_{\text{surf}} - T_{\text{in}}) - (T_{\text{surf}} - T_{\text{out}})] / \ln((T_{\text{surf}} - T_{\text{in}}) / (T_{\text{surf}} - T_{\text{out}}))$$

5. Heat Transfer Coefficient (h):

$$h = Q / (A \cdot \Delta T_{\text{LMTD}})$$

Analysis and Comparisons

1. Heat Transfer Rate (Q):

Forced convection results in significantly higher Q values compared to free convection, as airflow enhances heat dissipation. Pipe Bundle and Rectangular Fins demonstrate higher Q under free convection due to larger surface areas.

2. Efficiency (η):

Efficiency is highest for forced convection with the Pipe Bundle, highlighting the role of airflow in heat transfer.

3. Heat Transfer Coefficient (h):

h increases significantly with forced convection due to improved airflow and heat transfer rates.

These results illustrate the impact of geometry, convection type, and surface area on heat transfer performance

4. Log Mean Temperature Difference (LMTD):

LMTD is used to account for the temperature gradient between the surface and the air. It is highest for Fins in the natural convection case, reflecting the largest temperature gradient.

Comparison with Theory

For the forced convection scenario with the flat plate, the theoretical Nusselt number (Nu) is calculated using the empirical correlation for laminar flow over a flat plate:

$$Nu = 0.664 Re^{0.5} Pr^{1/3}$$

where Re is the Reynolds number, and Pr is the Prandtl number.

1. Reynolds Number (Re):

$$Re = (\rho v L) / \mu$$

Given:

- $\rho = 1.2 \text{ kg/m}^3$ (air density),
- $v = 2.91 \text{ m/s}$ (air velocity),
- $L = 0.2 \text{ m}$ (characteristic length of the flat plate),
- $\mu = 1.81 \times 10^{-5} \text{ Pa}\cdot\text{s}$ (dynamic viscosity).

Calculation:

$$\text{Re} = (1.2 \times 2.91 \times 0.2) / (1.81 \times 10^{-5}) = 38,556$$

2. Prandtl Number (Pr):

$$\text{Pr} = (\mu c_p) / k$$

Given:

- $\mu = 1.81 \times 10^{-5} \text{ Pa}\cdot\text{s}$,
- $c_p = 1005 \text{ J/(kg}\cdot\text{K)}$ (specific heat capacity of air),
- $k = 0.0262 \text{ W/(m}\cdot\text{K)}$ (thermal conductivity of air).

Calculation:

$$\text{Pr} = (1.81 \times 10^{-5} \times 1005) / 0.0262 = 0.695$$

3. Theoretical Nusselt Number (Nu):

$$\text{Nu} = 0.664 \text{ Re}^{0.5} \text{ Pr}^{1/3}$$

Given:

- $\text{Re} = 38,556$,
- $\text{Pr} = 0.695$.

Calculation:

$$\begin{aligned} \text{Nu} &= 0.664 \times (38,556)^{0.5} \times (0.695)^{1/3} \\ \text{Nu} &= 0.664 \times 196.3 \times 0.885 = 115.0 \end{aligned}$$

4. Experimental Nusselt Number (Nu):

$$\text{Nu} = (h L) / k$$

Given:

- $h = 112.31 \text{ W/(m}^2\cdot\text{K)}$ (experimental heat transfer coefficient),
- $L = 0.2 \text{ m}$,
- $k = 0.0262 \text{ W/(m}\cdot\text{K)}$.

Calculation:

$$\text{Nu} = (112.31 \times 0.2) / 0.0262 = 114.9$$

Comparison and Analysis

The experimental Nusselt number ($\text{Nu} = 114.9$) closely matches the theoretical value ($\text{Nu} = 115.0$). This agreement validates the experimental setup and measurement accuracy. Minor deviations can be attributed to:

- Heat losses to the surroundings,
- Non-idealities in the flow regime (e.g., turbulence or boundary layer effects),
- Measurement uncertainties in temperature, velocity, or power.

This comparison highlights the importance of forced convection in enhancing heat transfer rates and demonstrates the reliability of theoretical models for engineering applications. The results emphasize the role of flow type, geometry, and thermal properties in determining heat transfer efficiency.

Significance of Results

This analysis highlights the consistency of experimental and theoretical approaches in evaluating convective heat transfer. The forced convection scenario demonstrates the importance of flow type in enhancing heat transfer rates. The close agreement underscores the reliability of theoretical models and their applicability in practical heat transfer design and analysis. These findings emphasize that forced convection is crucial for achieving efficient thermal management in engineering applications.

3. Discussion

1. Key Questions Addressed

What are the differences between laminar and turbulent flows? Which has a higher heat transfer coefficient?

Laminar Flow: Characterized by smooth, orderly fluid motion in parallel layers with minimal mixing between layers. Heat transfer is primarily due to conduction, leading to a lower heat transfer coefficient.

Turbulent Flow: Involves chaotic fluid motion with significant mixing between layers, enhancing heat transfer due to increased convection. Turbulent flow has a much higher heat transfer coefficient than laminar flow.

In this experiment, forced convection scenarios demonstrated higher heat transfer coefficients due to turbulent or transitional flow induced by increased airflow velocity.

What is the difference in the measured values of the heat transfer coefficient when using linear average temperature instead of LMTD?

Using a linear average temperature instead of the Log Mean Temperature Difference (LMTD) can lead to inaccuracies because LMTD more accurately accounts for the exponential decay of the temperature gradient along the flow. In this experiment, LMTD was crucial for capturing the true driving force for heat transfer, particularly under varying inlet and outlet conditions. Using a linear average would underestimate the heat transfer coefficient.

2. Experimental Results vs. Theoretical Predictions

The comparison between experimental and theoretical Nusselt numbers for the forced convection flat plate scenario showed close agreement:

- Theoretical Nu: 115.0
- Experimental Nu: 114.9

This consistency validates the experimental approach and aligns with theoretical heat transfer models. Minor deviations could result from:

- Heat losses to the surroundings.
- Non-uniform flow conditions in the duct.
- Measurement inaccuracies (e.g., thermocouple placements or power calibration).

3. Sources of Error

1. Instrumentation Errors:

- Variations in thermocouple calibration may have introduced small errors in temperature measurements.
- Air velocity measurements might not reflect uniform flow conditions, particularly near duct walls.

2. Heat Losses:

- Heat losses due to conduction and radiation to the surroundings, especially in natural convection scenarios, might have affected the accuracy of Q and h .

3. Assumptions in Theoretical Models:

- Assumptions of ideal flow and uniform surface temperature might not hold, particularly in turbulent regimes or for geometries like pipe bundles.

4. Conclusion

The experiment on free and forced convection successfully demonstrated the principles of heat transfer through convection for various geometries (flat plate, pipe bundle, and fins). Key findings include:

1. Comparison of Free and Forced Convection:

- Forced convection consistently exhibited higher heat transfer rates and efficiencies due to enhanced fluid motion, which increases the convective heat transfer coefficient.
- Natural convection showed lower heat transfer rates, as it relies solely on buoyancy-driven flow.

2. Geometry's Influence:

- Larger surface areas, as seen in the pipe bundle and fins, significantly improved heat transfer performance, particularly in natural convection scenarios.

3. Alignment with Theory:

- Experimental results closely matched theoretical predictions for the Nusselt number, confirming the accuracy of the experiment and theoretical models.

This experiment highlighted the critical role of flow type, geometry, and surface area in determining heat transfer efficiency. It also validated the reliability of theoretical heat transfer models in predicting performance under real-world conditions.

5. Recommendations and Novel Ideas

Recommendations:

1. Enhanced Data Acquisition:

- Incorporate advanced data logging systems to continuously record temperature and flow rates. This reduces manual recording errors and allows for better time-dependent analysis of transient states.

2. Improved Insulation:

- Implement high-quality thermal insulation for the duct system and heated surfaces to minimize heat losses, thereby improving the precision of results.

3. Integration of Advanced Sensors:

- Use non-intrusive temperature and flow sensors, such as infrared thermography and laser Doppler anemometry, to provide more detailed and accurate measurements.

4. Dynamic Testing:

- Explore dynamic scenarios, such as varying flow velocities or surface temperatures over time, to analyze non-steady-state heat transfer behavior.

5. Flow Visualization:

- Using flow visualization techniques, such as smoke trails, would help validate flow regimes (laminar, transitional, or turbulent).

6. Precision Instruments:

- Calibrating thermocouples and using advanced flow meters could improve measurement reliability.

7. Expanded Geometries:

- Testing additional geometries and varying surface materials would provide broader insights into convection behavior.

Novel Ideas:

1. Variable Geometry Testing:

- Design adjustable test geometries, such as interchangeable plates with different roughness or perforations, to investigate their effects on heat transfer coefficients.

2. Multi-Fluid Analysis:

- Test the system with different working fluids (e.g., water, oils, or refrigerants) to explore the influence of fluid properties on convection.

3. 3D Flow Visualization:

- Use computational fluid dynamics (CFD) software to simulate and visualize flow patterns in three dimensions, providing deeper insights into flow behavior and its impact on heat transfer.

4. Energy Recovery System:

- Investigate the feasibility of incorporating heat recovery units to capture and reuse heat from the exhaust, showcasing practical energy efficiency improvements.

5. Environmental Impact Study:

- Extend the experiment to study the environmental implications of heat transfer systems, such as minimizing energy waste and optimizing system designs for sustainability.